Theorem 2(b): $x \cdot 0 = 0$

Proof:

$$
\begin{aligned}
x\cdot 0 &= x\cdot 0 + 0 \text{ (Postulate 2(a))}\\
&= x\cdot 0 + x\cdot x' \text{ (Postulate 5(b))}\\
&= x\cdot(0+x') \text{ (Postulate 4(a))}\\
&= x\cdot(x'+0) \text{ (Postulate 3(a))}\\
&= x\cdot x' \text{ (Postulate 2(a))}\\
&= 0 \text{ (Postulate 5(b))} \ \square
\end{aligned}
$$

Theorem 3: $(x')' = x$

Proof:

$$
\begin{aligned}
(x')' &= (x')' + 0 \text{ (Postulate 2(a))}\\
&= (x')' + (x')'\cdot(x')' \text{ (Postulate 5(b))}\\
&= ((x')'+x)\cdot((x')'+x') \text{ (Postulate 4(b))}\\
&= ((x')'+x)\cdot(x'+(x')') \text{ (Postulate 3(a))}\\
&= ((x')'+x)\cdot(1) \text{ (Postulate 5(a))}\\
&= ((x')'+x)\cdot(x+x') \text{ (Postulate 5(a))}\\
&= (x+(x')')\cdot(x+x') \text{ (Postulate 3(a))}\\
&= x+(x')'\cdot x' \text{ (Postulate 4(b))}\\
&= x+(x')\cdot(x')' \text{ (Postulate 3(b))}\\
&= x+0 \text{ (Postulate 5(b))}\\
&= x \text{ (Postulate 2(a))} \ \square
\end{aligned}
$$

The theorems involving two or three variables may be proven algebraically from the postulates and the theorems which have already been proven.

The Absorption theorem

Theorem 6(a): $x + xy = x$

Proof:

$$
\begin{aligned}
x+xy &= x\cdot 1 + xy \text{ (Postulate 2(b))}\\
&= x\cdot(1+y) \text{ (Postulate 4(a))}\\
&= x\cdot(y+1) \text{ (Postulate 3(a))}\\
&= x\cdot 1 \text{ (Theorem 2(a))}\\
&= x \text{ (Postulate 2(b))} \ \square
\end{aligned}
$$